

Notes: Kaustia & Torstila (JFE, 2011)

Stock Market Aversion? Political Preferences and Stock Market Participation

Contents

1	Big picture	3
2	Data and measurement (institutional context and key objects)	4
2.1	Political setting and the right-left measure	4
2.2	Zip-code-level stock ownership and election data	4
2.3	Zip-code controls and descriptive facts	5
2.4	Individual exit-poll sample	5
2.5	Members of parliament data	5
2.6	National poll and EVA values survey	6
2.7	Main dependent variables	6
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Conceptual framework and competing channels	7
3.2	Zip-code participation regression	7
3.3	Individual exit-poll logit	8
3.4	Members of parliament logit	8
3.5	Participation, allocation, and savings tests	8
3.6	Mechanism regressions for attitudes and expectations	9
3.7	Social-capital and trust auxiliary tests	9
3.8	Instrumental-variable attempt and robustness	9
4	Identification and empirical design (what makes the evidence credible)	10
4.1	What identifies the core evidence	10
4.2	Why geographic sorting is not enough	10
4.3	Why wealth and income are unlikely to be the whole story	10
4.4	Why reverse causality is limited	10
4.5	Why risk aversion is not the main channel	11
4.6	Why the mechanism evidence favors values over expectations	11
4.7	Why social capital and trust do not explain the results	11
4.8	What the paper can and cannot pin down	12
5	Tables and figures	12
5.1	Table 1: Descriptive statistics	12

5.2	Table 2: Zip-code-level participation regressions	12
5.3	Table 3: Individual exit-poll regressions	13
5.4	Table 4: Members of parliament	13
5.5	Table 5: Participation versus allocation and savings	15
5.6	Table 6: Value-expressive attitudes, safety-net expectations, and return expectations	16
6	Short summary	16
7	Conclusion	16
7.1	Core conclusion	16
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** Are people with left-wing political preferences less likely to participate in the stock market, even after controlling for income, wealth, education, and other standard determinants of stock ownership?
- **Core idea.** The paper treats political preference as a proxy for **personal values**. If some households see the stock market as inconsistent with their values, then stock ownership carries a psychological participation cost. The paper calls this mechanism **stock market aversion**.
- **Why Finland is useful.** Finland offers unusually good data for this question: official registry data on stock ownership, detailed election data, a multiparty system that creates meaningful left-right variation, and public information on politicians' income and wealth.
- **Main data.** The paper triangulates the result with **four data sources**: zip-code-level registry and election data, an individual exit poll, a sample of members of parliament (MPs), and a nationally representative survey plus an auxiliary values-and-attitudes survey.
- **Main empirical design.** Regress stock market participation on a right-left political measure, first at the zip-code level and then at the individual level, while controlling for income, wealth, education, housing, age, voting activity, and fixed effects where available.
- **Baseline zip-code result.** A one-point move to the left on the paper's 1–10 political scale is associated with about a **6% lower** stock-market participation rate. A move from a moderate right party to a moderate left party lowers participation by about **20%**.
- **Individual result.** In the exit-poll data, a one-point move to the left is associated with a **5.8% lower** likelihood of owning stocks or equity mutual funds. The effect is comparable in size to a substantial drop in income.
- **Elite-sample result.** Among Finnish MPs, right-wing politicians are much more likely to own stock than left-wing politicians. The raw ownership rates are about **51%** for right-wing MPs versus **21%** for left-wing MPs.
- **Mechanism result.** Political orientation predicts the **decision to participate**, but not the **equity allocation conditional on participation**. Left-wing respondents are also much more likely to say that the stock market is socially harmful, while they do *not* differ much in expected stock returns.
- **Main credibility result.** The pattern survives controls for income, wealth, education, home ownership, social-capital proxies, polling-station fixed effects, and a battery of robustness checks. It also appears in groups that are clearly above the usual participation threshold, such as MPs.
- **Economic takeaway.** The paper argues that limited stock-market participation is not only about wealth, information, or risk tolerance. **Values and identity can matter directly for whether households enter the market at all.**

2 Data and measurement (institutional context and key objects)

2.1 Political setting and the right-left measure

- Finland elects a **200-seat parliament** every four years under a **d'Hondt proportional-representation** system.
- The multiparty structure is useful because parties occupy different points on a meaningful **left-right ideological scale**, rather than forcing all political variation into a two-party split.
- The paper uses the party scores from **Hix and Lord (1997)** and linearly transforms them into a **1.0 to 10.0 right-left axis**, where **higher values mean more left-wing politics**.
- In area-level analysis, the political variable is the **vote-share-weighted average** of party scores. As an alternative, the paper also uses the **share of votes cast for parties with score above 5.0**.

Interpretation. Political orientation is the paper's empirical proxy for values. The point is not that politics causes investing mechanically, but that ideology summarizes broader attitudes toward markets, capitalism, and redistribution.

2.2 Zip-code-level stock ownership and election data

- Stock ownership comes from the official registry of the **Finnish Central Securities Depository (FCSD)** covering **January 1995 to November 2002**.
- Parliamentary elections occurred in **1995, 1999, and 2003**. Because the ownership registry ends before the 2003 election, the paper uses **November 2002 ownership** as a proxy for ownership in March 2003.
- The key area-level ownership measure is the **number of stock-market investors per population** in each zip-code area.
- Votes are observed at the **voting-precinct** level and then prorated into zip-code areas using the voting-age population in overlapping geographic units.
- For 1995 and 1999 the vote data cover **election-day votes only**; for 2003 the authors can also check whether excluding postal votes matters, and it does not materially change the results.
- After exclusions, the final pooled zip-code sample contains **6,322 observations** across the three election years.

Interpretation. This is the paper's cleanest large-sample test. The FCSD data are administrative, not self-reported, which avoids the usual survey nonresponse and misreporting problems.

2.3 Zip-code controls and descriptive facts

- The area-level controls include **median income**, **mean taxable wealth**, **share of high-income individuals**, **share of owner-occupied dwellings**, **education shares**, **age composition**, **share of Swedish speakers**, and **population density**.
- Continuous controls are log transformed, and population density enters through orthogonal-polynomial terms.
- In the pooled sample, the mean investor share is **9.8%** of the voting-age population.
- The mean election-day voting rate is **41.2%**, but this excludes postal ballots and therefore understates full turnout.
- The three major parties together account for about **70.6%** of votes on average.
- Stock participation rises over time in the descriptive statistics, from **7.9% in 1995** to **12.3% in 2003**.

Interpretation. The descriptive tables show large cross-sectional variation in both political orientation and participation. The paper's regression problem is therefore not lack of signal, but whether that signal is really about values rather than socioeconomic composition.

2.4 Individual exit-poll sample

- The exit poll was run in the Finnish parliamentary election of **March 16, 2003**.
- It covers **nine Helsinki voting precincts** deliberately chosen to span different socioeconomic environments.
- The survey yields **824 usable responses**.
- The ownership measure here is broader than in the zip-code data: the **investor dummy** includes **both direct stock ownership and equity mutual funds**.
- The questionnaire was kept short to keep refusal rates low, and the authors report a very low refusal rate.

Interpretation. This data set is important because it moves from area-level correlations to individual-level behavior. It is the paper's main answer to the claim that the zip-code evidence is driven only by neighborhood sorting.

2.5 Members of parliament data

- The Finnish Parliament has **200 MPs**; the paper obtains complete data for **191** of them as of **September 2006**.
- Stock ownership is collected from the FCSO public terminal.
- Taxable **income** and **wealth** are public in Finland and come from **Veroporssi**.

- Party affiliation, age, and education come from official Parliament sources.
- The authors exclude local phone-company certificate conversions because those do not reflect a voluntary stock-market investment decision.

Interpretation. MPs form a high-information, high-income group that should be well above standard participation thresholds. If ideology still matters here, the result is harder to explain with fixed costs or simple lack of financial awareness.

2.6 National poll and EVA values survey

- The national poll is a proprietary **TNS Gallup** panel survey conducted in **week 23 of 2009**.
- The panel contains about **1,300 respondents** and has a typical response rate of around **80%**.
- A key design advantage is that political views and voting intentions are already collected regularly, so respondents do not anticipate that investment questions will later be matched with political information.
- The paper also uses the **Finnish Business and Policy Forum (EVA) Survey on Finnish Values and Attitudes 2000**.
- The EVA survey is a mail survey sent to a random sample of **4,500** mainland Finns ages 18–70, with a response rate of about **50%**.

Interpretation. These surveys are used mainly for mechanism tests: attitudes toward the stock market, savings behavior, pension expectations, and return expectations.

2.7 Main dependent variables

- **Zip-code participation:** log number of stock-market investors per population.
- **Individual participation:** investor dummy equal to one for owning stocks or equity mutual funds.
- **MP participation:** investor dummy equal to one for owning listed Finnish stocks.
- **Conditional equity allocation:** share of wealth held in equity among those who participate.
- **Savings rate proxy:** ordered response to whether the respondent can save money monthly after regular expenses.
- **Value-expressive attitudes:** responses to questions such as whether the stock market is beneficial to society, excessively worshipped, good for the economy, or a sign of too much selfish profit seeking.
- **Alternative-channel variables:** expectations about future pensions, expected annual stock returns, and whether the current stock-market boom is a bubble.

Interpretation. The paper's structure is deliberate: first show the participation fact, then show that the fact is strongest where a values interpretation predicts it should be strongest.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework and competing channels

The paper studies three channels from political preferences to investing:

- **Value-expressive choice.** Left-wing voters may see the stock market as morally unattractive or socially harmful. This raises a *cognitive participation cost* and lowers entry into the market.
- **Safety-net expectations.** Left-wing voters may expect a stronger public pension system, so they save less privately and therefore participate less in equity markets.
- **Return expectations.** Left-wing voters may simply expect lower stock returns.

These channels make different empirical predictions. Under the value-expressive view, politics should mainly affect **whether** someone participates, not **how much equity** they hold once they are already in the market. Under the safety-net view, politics should also show up in **savings behavior**. Under the return-expectations view, politics should show up in **expected returns** and perhaps in **conditional equity allocation**.

Interpretation. This prediction structure is a major strength of the paper. The mechanism section is not just an add-on; it is what lets the authors discriminate between values, savings motives, and beliefs about future returns.

3.2 Zip-code participation regression

The paper’s main area-level specification can be summarized as

$$Investors_{zt} = \alpha + \beta PoliticalLeft_{zt} + X'_{zt}\gamma + \delta_t + \varepsilon_{zt}, \quad (1)$$

where:

- $Investors_{zt}$ is the log number of stock-market investors per population in zip code z at election t ,
- $PoliticalLeft_{zt}$ is either the weighted average right-left score or the left-wing vote share,
- X_{zt} includes votes per population, income, wealth, high-income share, homeownership, education, demographic controls, and population-density controls,
- δ_t are election fixed effects.

Standard errors are heteroskedasticity-robust and clustered at the **zip-code level**.

Interpretation. Because the dependent variable is logged, the coefficient on political orientation has an approximate percentage interpretation. A negative β means more left-leaning areas have lower stock-market participation, conditional on observable socioeconomic composition.

3.3 Individual exit-poll logit

At the individual level the paper estimates

$$\Pr(Investor_i = 1) = \Lambda(\alpha + \beta PoliticalLeft_i + X_i' \gamma + \phi_{p(i)}), \quad (2)$$

where $Investor_i$ indicates stock or equity-mutual-fund ownership, X_i includes sex, age dummies, and income dummies, and $\phi_{p(i)}$ are **polling-station fixed effects**.

Interpretation. This is the paper's cleanest individual-level specification. The polling-station effects mean the comparison is partly within very local geographic areas rather than only across different neighborhoods.

3.4 Members of parliament logit

For MPs the empirical model is

$$\begin{aligned} \Pr(Investor_m = 1) = \Lambda & \left(\alpha + \beta PoliticalLeft_m + \gamma_1 \log Income_m + \gamma_2 \log Wealth_m \right. \\ & \left. + \gamma_3 Male_m + \gamma_4 Age_m + \gamma_5 Age_m^2 + \gamma_6 College_m \right). \end{aligned} \quad (3)$$

Interpretation. This test asks whether ideology still matters in a group that is financially sophisticated, politically informed, and almost certainly above the normal participation threshold. A negative β here is especially hard to square with simple fixed-cost stories.

3.5 Participation, allocation, and savings tests

The national-poll data allow the paper to separate the participation margin from the portfolio-choice margin. The participation equation is another logit of the form

$$\Pr(Investor_i = 1) = \Lambda(\alpha + \beta PoliticalLeft_i + X_i' \gamma). \quad (4)$$

For the conditional equity share the paper estimates a Tobit-style bounded-allocation model,

$$EquityShare_i = \alpha + \beta PoliticalLeft_i + X_i' \gamma + \varepsilon_i, \quad (5)$$

where the dependent variable is the **percentage of wealth held in equity among participants**.

For the savings-rate proxy the paper estimates an ordered-probit model,

$$SavingsCategory_i^* = \alpha + \beta PoliticalLeft_i + X_i' \gamma + \varepsilon_i. \quad (6)$$

Interpretation. If politics works through a cognitive participation cost, then β should be negative in the participation equation but close to zero in the conditional-allocation equation. If politics works through social-safety-net beliefs, then β should also show up in savings behavior.

3.6 Mechanism regressions for attitudes and expectations

For the mechanism tests the paper estimates ordered probits or OLS of the general form

$$Y_i = \alpha + \beta PoliticalLeft_i + X_i' \gamma + \varepsilon_i, \quad (7)$$

where Y_i is one of the following:

- whether the stock market is beneficial or harmful to society,
- whether the stock market is excessively worshipped,
- whether stock investment is good for the economy,
- whether society has succumbed too much to selfish profit seeking,
- whether pensions will get worse,
- expected annual stock returns,
- whether the current boom is not a bubble.

Interpretation. This battery of outcomes is central. The paper is not satisfied with showing a reduced-form participation correlation; it directly asks which beliefs and attitudes move with ideology.

3.7 Social-capital and trust auxiliary tests

To assess whether social capital could explain the main result, the paper replicates the spirit of **Glaeser, Laibson, Scheinkman, and Soutter (2000)** using **World Values Survey** data for Scandinavian countries. It augments standard social-capital regressions with political variables and studies:

- the number of organizations to which the respondent belongs,
- and a trust dummy for whether “other people can be trusted.”

Interpretation. This is not the paper’s main design, but it matters because prior household-finance work links trust and social capital to stock participation. If left-wing ideology were just a disguised trust variable, the main result would be much less interesting.

3.8 Instrumental-variable attempt and robustness

The paper also explores IV versions of the zip-code regressions, instrumenting political variables with demographic features such as the **share of Swedish speakers**, **population density**, and **occupational composition**. The IV coefficients remain negative, but the authors conclude that the instruments are **weak** and therefore do not emphasize those estimates.

Interpretation. This is the right attitude. The paper’s contribution comes mainly from cross-dataset consistency and mechanism evidence, not from a heroic causal-IV claim.

4 Identification and empirical design (what makes the evidence credible)

4.1 What identifies the core evidence

The core empirical fact is that more left-leaning voters, areas, and politicians are less likely to own stock. The identifying variation comes from differences in political orientation **conditional on rich socioeconomic controls**.

Interpretation. This is not a natural experiment. The paper is persuasive because it repeatedly finds the same pattern in very different samples, not because any single regression fully settles causality.

4.2 Why geographic sorting is not enough

A pure neighborhood-sorting story would say that rich right-wing investors simply live in different places from poorer left-wing noninvestors. The paper addresses this in two ways:

- it controls extensively for area-level income, wealth, housing, education, and density;
- it reproduces the result in **individual-level exit-poll data** with **polling-station fixed effects**.

Interpretation. Once the result shows up within local polling areas and at the individual level, it is much harder to dismiss the zip-code evidence as purely ecological.

4.3 Why wealth and income are unlikely to be the whole story

The paper takes the wealth-confound seriously.

- Zip-code regressions include income, wealth, high-income share, and home ownership.
- The authors also test nonlinear controls and income-quartile splits.
- The individual exit-poll regressions include detailed income dummies.
- The MP regressions include both income and taxable wealth.

Interpretation. Wealth clearly matters for participation, but the political variable has incremental explanatory power even after the obvious wealth channels are controlled for.

4.4 Why reverse causality is limited

Could investing itself make people more right-wing? The paper argues that this is unlikely to explain the main findings.

- Political preferences are thought to form relatively early in life.

- MPs have highly visible and explicit ideological affiliations, making their political type particularly clear.
- In the national Gallup survey, political information is collected independently of the special investment questions.

Interpretation. Reverse causality cannot be ruled out completely in an observational paper, but it is hard to believe that it drives the entire result across all four data sets.

4.5 Why risk aversion is not the main channel

A simple risk-aversion story predicts lower stock participation for more risk-averse people, but it also tends to predict lower **equity allocation conditional on participation**. The paper does not find that second pattern.

Interpretation. The absence of a conditional-allocation effect is one of the most important pieces of evidence in the paper. Politics seems to matter like a fixed participation cost, not like a standard curvature parameter in utility.

4.6 Why the mechanism evidence favors values over expectations

The value-expressive view predicts that ideology should strongly affect **attitudes toward the stock market**. The safety-net story predicts differences in **savings or pension beliefs**. The return-expectation story predicts differences in **expected stock returns**.

The data line up most strongly with the first channel:

- left-wing respondents are much more anti-stock-market in attitude questions,
- left-wing respondents are *not* meaningfully more pessimistic about future returns,
- and they are not more optimistic about future pensions.

Interpretation. This is the paper's strongest substantive claim. The mechanism evidence points much more toward *values and identity* than toward *beliefs about payoffs*.

4.7 Why social capital and trust do not explain the results

Prior work shows that civic engagement and trust help explain stock participation. The paper therefore controls for voting activity and home ownership in zip-code regressions, and also uses individual-level survey evidence on trust and organization membership.

Interpretation. The social-capital story helps explain why voting activity predicts participation, but it does *not* explain why left-wing orientation itself predicts lower participation. In the survey evidence, left-wing views are not negatively related to trust.

4.8 What the paper can and cannot pin down

What the paper does well:

- documents a robust new predictor of stock-market participation,
- shows that the effect is economically large,
- and narrows the plausible transmission mechanism.

What the paper does *not* fully establish:

- a sharp causal estimate from exogenous political shocks,
- whether MP behavior reflects personal ideology or strategic concern for voters,
- and whether the same magnitudes carry over outside Finland.

Interpretation. The paper is strongest as a **careful empirical demonstration that tastes matter in household finance**, not as a final causal decomposition of why those tastes arise.

5 Tables and figures

5.1 Table 1: Descriptive statistics

Read.

- In the pooled zip-code sample, the mean investor share is **9.8%** of the voting-age population.
- The mean election-day voting rate is **41.2%**, with higher full turnout once postal ballots are included.
- The three biggest parties account for about **70.6%** of votes.
- The mean investor share rises from **7.9% in 1995** to **12.3% in 2003**.
- There is substantial geographic dispersion in party vote shares and investor participation.

Economic message. The paper has enough cross-sectional variation to identify meaningful participation differences across political environments.

5.2 Table 2: Zip-code-level participation regressions

Read.

- The coefficient on the right-left axis is **-0.129** without controls, **-0.060** with core controls, and **-0.051** with the fullest specification.
- Using left-wing vote share instead of the continuous axis gives similarly strong negative coefficients, such as **-0.428** in the fullest specification.

Table 1
Descriptive statistics on zip-code-level investor and voting data.
The sample consists on votes by zip code area in the Finnish parliamentary elections of 1995, 1999, and 2003. Panel A shows zip-code-level descriptive statistics for the voting age (over 18 years) population, percentage of voters in the population, percentage of investors in the population, and the vote shares of the three main parties separately and aggregated. Panel B shows the aggregate numbers for the entire country. The data are based on votes given on election day, i.e., they exclude postal votes given before the election day.

				% share of votes			
Year	Voting age population	% voting ^a	% investors	Social Democrats	Center Party	Conservatives	Total, big three
Panel A: Unit of analysis=zip code area							
Mean 1995	1,688	40.2	7.9	22.9	33.1	13.8	69.8
Median	693	40.5	7.2	22.4	32.0	13.0	72.8
St. dev.	2,190	9.6	4.6	12.5	23.1	9.0	16.0
Min	95	3.7	0.6	0.7	0.5	0.2	4.5
Max	18,316	98.8	40.0	75.4	92.8	64.2	98.3
N	2,196	2,196	2,196	2,196	2,196	2,196	2,196
Mean 1999	1,836	40.5	9.4	19.5	34.3	16.3	70.1
Median	783	40.6	8.6	18.7	33.3	15.3	72.7
St. dev.	2,395	8.7	4.9	11.1	22.7	10.1	16.3
Min	92	11.6	0.9	0.5	0.4	0.2	3.0
Max	20,595	93.5	40.6	79.5	90.5	55.7	98.5
N	2,087	2,087	2,087	2,087	2,087	2,087	2,087
Mean 2003	1,698	43.0	12.3	20.7	36.9	14.2	71.8
Median	717	42.5	11.2	20.0	36.0	12.9	74.6
St. dev.	2,238	10.4	5.9	11.2	22.2	9.3	16.6
Min	58	2.0	1.5	0.6	0.3	0.1	1.8
Max	21,074	93.1	65.7	73.9	90.2	55.9	98.6
N	2,039	2,039	2,039	2,039	2,039	2,039	2,039
Mean Pooled	1,740	41.2	9.8	21.1	34.7	14.7	70.6
Median	727	41.2	8.8	20.3	33.8	13.8	73.5
St. dev.	2,275	9.7	5.5	11.7	22.8	9.5	16.3
Min	58	2.0	0.6	0.5	0.3	0.1	1.8
Max	21,074	98.8	65.7	79.5	92.8	64.2	98.6
N	6,322	6,322	6,322	6,322	6,322	6,322	6,322
Panel B: Unit of analysis=entire country							
1995	3,705,796	40.0	12.4	28.3	17.7	18.4	46.0
1999	3,830,696	39.7	13.9	23.3	19.7	21.7	42.9
2003	3,462,869	42.3	17.5	24.9	23.8	18.9	48.7
Mean	3,666,454	40.7	14.6	25.5	20.4	19.6	45.9

^a % voting is based on election day only and thus excludes postal ballots.

Table 2
Determinants of stock ownership in zip-code-level data.
Specifications 1 through 6 are OLS. Zip-code-level observations for three Finnish parliamentary elections (1995, 1999, and 2003) are pooled together. Investors is the log number of stock market investors per population. Right-left axis value is a weighted-average political orientation of the zip code, based on party-by-party score on a scale of 1.0 (right) to (-1.0) (left). Source: Hix and Lord, 1997. Left-wing proportion is the number of votes for parties with a right-left score of more than 5.0, divided by the total number of votes given. Votes is the log number of votes given per population. Income is the log of median income of individuals. Wealth is the log mean taxable wealth of families. High income individuals is the log number of people in the highest income bracket per population. House owners is the log number of owner-occupied dwellings in the area per population. College degrees is the log number of people with higher academic education per population. Heteroskedasticity and clustering (at the zip-code-level) corrected t-statistics are below coefficients in parentheses. All significance tests are two-sided. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Dependent variable: Investors					
	(1)	(2)	(3)	(4)	(5)	(6)
Right-left axis value	-0.129*** (-8.59)	-0.060*** (-4.52)	-0.051*** (-4.29)			
Left-wing proportion				-0.540*** (-9.12)	-0.247*** (-4.40)	-0.428*** (-8.19)
Votes	0.508*** (7.52)	0.126** (2.00)	0.231*** (3.91)	0.461*** (6.72)	0.101 (1.58)	0.184*** (3.31)
Income		0.293*** (7.41)	0.174*** (2.97)		0.317*** (7.75)	0.248*** (4.24)
Wealth		0.518*** (21.84)	0.337*** (16.65)		0.509*** (20.62)	0.295*** (14.41)
High income individuals			0.230*** (8.65)			0.237*** (9.07)
House owners			0.373*** (6.20)			0.368*** (6.20)
College degrees			0.156*** (8.77)			0.150*** (8.51)
Election fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Population density controls	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls	No	No	Yes	Yes	Yes	Yes
N	6,322	6,322	6,322	6,322	6,322	6,322
R-squared	0.31	0.48	0.62	0.31	0.48	0.62

- The political coefficients remain significant after adding income, wealth, high-income share, home ownership, education, election fixed effects, density controls, and demographic controls.
- The paper interprets the estimates as roughly a **6.2%** lower participation rate for each one-point move left on the political scale.
- A move from the main right-wing party to the main left-wing party implies about a **19.8%** lower participation rate.

Economic message. The main fact is already visible in the broad administrative data: more left-leaning areas participate less in the stock market, even after extensive controls.

5.3 Table 3: Individual exit-poll regressions

Read.

- The right-left coefficient in the logit is **-0.23** with a t-statistic of about **-4.55**.
- Replacing the continuous ideology measure with a left-wing vote dummy gives a coefficient of about **-0.68** with a t-statistic of about **-4.22**.
- The paper interprets the continuous coefficient as a **5.8% lower** probability of market participation for a one-point move left.
- The magnitude is comparable to a meaningful drop in income.

Economic message. The result survives the move from neighborhood averages to actual individual voters, and it also survives when mutual-fund ownership is included.

5.4 Table 4: Members of parliament

Read.

Table 3

Determinants of stock ownership in exit poll data.

Both specifications are logit. The dependent variable used is a dummy for owning stock as of the election date in the 2003 Finnish parliamentary election. Right-left axis value is a score of 1.0 to 10.0 for Finnish parties based on [Hix and Lord \(1997\)](#), with higher values representing left-wing parties. The left-wing vote dummy takes a value of one for voting any of the left-wing parties with a [Hix and Lord \(1997\)](#) right-left score above 5.0. The reference group for age dummies is 36–45 years and for income dummies 2,000–2,999 euros. Heteroskedasticity corrected t-statistics are in parentheses below the coefficients. All significance tests are two-sided. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Dependent variable: Investor dummy	
	(1)	(2)
Right-left axis value	–0.23*** (–4.55)	
Left-wing vote dummy		–0.68*** (–4.22)
Male dummy	–0.02 (–0.16)	–0.007 (–0.04)
Age 18–25 dummy	0.22 (0.73)	0.23 (0.77)
Age 26–35 dummy	0.08 (0.38)	0.08 (0.38)
Age 46–55 dummy	0.13 (0.47)	0.14 (0.55)
Age 56–65 dummy	0.66** (2.14)	0.73** (2.40)
Age 66 or above dummy	0.11 (0.29)	0.17 (0.45)
Income 0–999 euros dummy	–0.35 (–1.31)	–0.36 (–1.39)
Income 1,000–1,999 euros dummy	–0.85*** (–3.70)	–0.90*** (–3.97)
Income 3,000–3,999 euros dummy	0.82*** (3.31)	0.76*** (3.12)
Income 4,000–4,999 euros dummy	0.64** (2.13)	0.653** (2.18)
Income 5,000 euros or above dummy	1.24*** (3.80)	1.27*** (3.94)
Constant	1.71*** (4.01)	0.81** (2.51)
Polling station fixed effects	Yes	Yes
N	782	806
Pseudo R-squared	0.12	0.11

Table 4

Determinants of stock ownership for members of parliament.

Both specifications are logits in which the dependent variable (Investor dummy) takes the value of one if the individual member of the Finnish parliament owns stocks as of September 2006, and zero otherwise. Right-left axis value is a score of 1.0 to 10.0 for Finnish parties based on [Hix and Lord \(1997\)](#), with higher values representing left-wing parties. The left-wing dummy takes a value of one for belonging to any of the left-wing parties with a [Hix and Lord \(1997\)](#) right-left score above 5.0. Log income and log wealth are calculated as the natural logarithm of taxable income and taxable wealth, in euros, in the 2004 tax year. The parliament has 200 members; income and wealth data are missing for nine individuals. Heteroskedasticity corrected t-statistics are in parentheses below the coefficients. All significance tests are two-sided. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	Dependent variable: Investor dummy	
	(1)	(2)
Right-left axis	–0.209** (–2.04)	
Left-wing dummy		–0.721** (–2.21)
Log income	0.828 (1.45)	0.832 (1.45)
Log wealth	0.076** (2.50)	0.075** (2.47)
Male dummy	0.520 (1.31)	0.493 (1.23)
Age	0.065 (0.34)	0.059 (0.31)
Age squared	0.000 (0.23)	0.000 (0.20)
College degree dummy	0.017 (0.05)	0.022 (0.06)
Constant	–13.547 (–1.53)	–11.945 (–1.34)
N	191	191
Pseudo R-squared	0.11	0.11

- Raw ownership is much higher for right-wing MPs than left-wing MPs: roughly **51% versus 21%**.
- In the controlled logit, the coefficient on right-left ideology is about **-0.209**; with a left-wing dummy it is about **-0.721**.
- **Log wealth** enters positively and significantly, while log income is positive but not statistically strong.
- The paper interprets a three-point move from moderate right to moderate left as about a **17.5%** lower likelihood of stock ownership.

Economic message. Even among financially sophisticated political elites, ideology predicts stock ownership. This is hard to reconcile with a story based only on ignorance or low wealth.

5.5 Table 5: Participation versus allocation and savings

Read.

- In the national Gallup survey, ideology again predicts participation: the right-left coefficient is about **-0.166**.
- Conditional on participating, the coefficients on equity allocation are approximately **-0.002** and **0.012**, both economically tiny and statistically insignificant.
- For savings behavior, the ordered-probit coefficient is about **-0.082**; in the Swedish survey it is about **-0.068**.

Economic message. Politics matters at the entry margin, not the portfolio-share margin. There is some evidence that left-wing respondents save less frequently, but the much sharper pattern is the participation effect.

Table 5
Equity allocation and savings rates.

This table shows a comparative analysis of three alternative dependent variables. The data for Specifications 1, 2, and 4 come from a nationally representative poll conducted in Finland in week 23 of 2009; for Specification 3, from data on Finnish members of parliament as of September 2006; and for Specification 5, from the European Values Survey for Sweden for 1999. The first specification uses a dummy for stock market participation as a dependent variable; the second and third specifications use equity allocation conditional on stock market participation as a continuous variable from zero to 100%; the fourth specification is a variable taking the values from 1–5 for an answer to the question “Are you able to save money monthly after regular expenses never/rarely/sometimes/mostly/always?”; and the fifth specification a dummy variable for a positive answer to the question “Were you able to put some money aside last year?”. Right-left axis value is a score of 1.0 to 10.0 for Finnish parties based on Hix and Lord (1997), with higher values representing left-wing parties. Heteroskedasticity corrected *t*-statistics are below coefficients. All significance tests are two-sided. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1) Participation	(2) Equity allocation	(3) Equity allocation	(4) Savings rate	(5) Savings rate
	Logit Decision to participate	Tobit Equity allocation (%)	Tobit Equity allocation (%)	Ordered probit Higher monthly savings	Logit Able to save the year before
Right-left axis value	-0.166*** (-2.64)	-0.002 (-0.18)	0.012 (0.12)	-0.082*** (-2.88)	-0.068* (-1.92)
Male dummy	0.405** (2.07)	0.048 (1.58)	0.293 (0.62)	-0.112 (-1.30)	-0.071 (-0.44)
College degree dummy	0.237 (0.89)	-0.003 (-0.06)	0.339 (0.83)	0.291** (2.39)	0.155 (1.08)
Log income			-0.131 (-0.31)		
Income dummies	Yes	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	Yes	Yes	Yes
<i>N</i>	536	210	65	649	958
Source:	Gallup survey	Gallup survey	MP data	Gallup survey	Swedish survey

Table 6
Value-expressive choices, safety net expectations, and return expectations.

This table shows a comparative analysis of three alternative transmission mechanisms that may account for the association between stock market participation and political preferences. The data for Specifications 1, 5, and 6 come from a nationally representative poll conducted in Finland in week 23, 2009; for Specifications 2, 3, 4, and 7 from The Finnish Business and Policy Forum (FVA) survey on Finnish Values and Attitudes 2000. The dependent variables are specified as follows. Column 1: coded from 1–7 (very harmful to very beneficial) based on “Do you consider the stock market beneficial or harmful to society as a whole?” Column 2: coded from 1–5 (strongly disagree to strongly agree) with the statement “The stock market is not excessively worshipped in Finland.” Column 3: coded from 1–5 (strongly disagree to strongly agree) with the statement “Investment in shares is an excellent way to increase national wealth.” Column 4: coded from 1–5 (strongly disagree to strongly agree) with the statement “Finnish society has not succumbed too much to market forces and selfish profit seeking.” Column 5: coded from 1–7 (will get significantly better/worse) for the question “How do you think pensions will develop within the next 20 years?” Column 6: a continuous variable for expected annual stock returns over the next 20 years. Column 7: coded from 1–5 (strongly disagree to strongly agree) with the statement “The present stock market boom is not a bubble.” Right-left axis value is a score of 1.0 to 10.0 for Finnish parties based on Hix and Lord (1997), with higher values representing left-wing parties. Heteroskedasticity corrected *t*-statistics are below coefficients. All significance tests are two-sided. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

	(1) Value-expressive opinions about the stock market	(2) Value-expressive opinions about the stock market	(3) Stock market	(4) Safety net exp.	(5) Safety net exp.	(6) Return expectations	(7) Return expectations
	Ordered probit Stock market is beneficial	Ordered probit Stock market not excessively worshipped	Ordered probit Stock investments good for economy	Ordered probit Net too much selfish profit seeking	Ordered probit Pensions will get worse	OLS Expected stock return	Ordered probit Present boom is not a bubble
Right-left axis value	-0.222*** (-7.23)	-0.137*** (-6.72)	-0.103*** (-5.09)	-0.127*** (-6.20)	0.054* (1.92)	-0.534 (-0.55)	-0.021 (-1.05)
Male dummy	-0.079 (-0.87)	0.012 (0.19)	0.007 (0.12)	0.061 (0.99)	0.0001 (0.00)	1.465 (1.33)	0.101* (1.67)
College degree	-0.054 (-0.50)	0.197*** (3.07)	-0.042 (-0.66)	0.198*** (3.11)	-0.037 (-0.36)	-0.716 (-1.28)	0.193*** (2.97)
Income dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class dummies		Yes	Yes	Yes			Yes
Age dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	533	1,297	1,304	1,301	624	269	1,296
Data source:	Gallup survey	EVA survey	EVA survey	EVA survey	Gallup survey	Gallup survey	EVA survey

5.6 Table 6: Value-expressive attitudes, safety-net expectations, and return expectations

Read.

- The right-left coefficient is strongly negative in all four pro-stock attitude regressions: about **-0.222**, **-0.137**, **-0.103**, and **-0.127**.
- For pension expectations, the coefficient is actually slightly **positive** (about **0.054**), meaning left-wing respondents are if anything a bit more pessimistic about pensions, not more optimistic.
- For expected annual stock returns, the coefficient is about **-0.534** and statistically insignificant.
- For the “present boom is not a bubble” question, the coefficient is also insignificant.
- Descriptively, only about **27.7%** of those who say the stock market is harmful invest in stocks, versus about **57.9%** of those who say it is beneficial.

Economic message. The strongest channel is anti-stock-market sentiment, not lower return expectations and not stronger faith in the welfare state.

6 Short summary

- The paper shows that **left-wing political preference predicts lower stock-market participation** in Finland.
- The result appears in **zip-code administrative data**, an **individual exit poll**, a sample of **members of parliament**, and a **nationally representative survey**.
- The magnitude is large: a one-point move left on a 1–10 scale lowers participation by about **5–6%**; a moderate-right to moderate-left move lowers it by about **17–20%**.
- Political preference predicts **participation**, but not **equity allocation conditional on participation**.
- Left-wing respondents are much more likely to see the stock market as **socially harmful**, but they do not have meaningfully lower **expected stock returns**.
- The evidence is most consistent with a **value-expressive or identity-based participation cost**.

7 Conclusion

7.1 Core conclusion

The paper’s central claim is that stock-market participation depends not only on wealth, education, and risk, but also on **personal values**. Political ideology serves as a measurable proxy for those values, and more left-wing individuals are systematically less likely to own stocks.

7.2 Economic intuition

The most natural interpretation is that some households treat the stock market partly like a **consumption good with identity content**. Holding stock is not just a bet on cash flows; it can also feel like endorsement of a market institution. If that institution clashes with a person's values, the person may choose nonparticipation.

7.3 Takeaway

For household finance, the paper suggests that limited participation is partly a **tastes and identity problem**. For political economy, it suggests that ideology shapes not only voting and consumption, but also one of the most important household financial decisions: whether to enter the stock market at all.